

Solution By equality of two matrices, equating the corresponding elements, we get

$$2a + b = 4 \qquad 5c - d = 11$$

$$a - 2b = -3 \qquad 4c + 3d = 24$$

Solving these equations, we get

$$a = 1, b = 2, c = 3 \text{ and } d = 4$$

EXERCISE 3.1

1. In the matrix $A = \begin{bmatrix} 2 & 5 & 19 & -7 \\ 35 & -2 & \frac{5}{2} & 12 \\ \sqrt{3} & 1 & -5 & 17 \end{bmatrix}$, write:

- (i) The order of the matrix, (ii) The number of elements,
- (iii) Write the elements a_{13} , a_{21} , a_{33} , a_{24} , a_{23} .
2. If a matrix has 24 elements, what are the possible orders it can have? What, if it has 13 elements?
3. If a matrix has 18 elements, what are the possible orders it can have? What, if it has 5 elements?
4. Construct a 2×2 matrix, $A = [a_{ij}]$, whose elements are given by:

$$(i) \ a_{ij} = \frac{(i+j)^2}{2} \qquad (ii) \ a_{ij} = \frac{i}{j} \qquad (iii) \ a_{ij} = \frac{(i+2j)^2}{2}$$

5. Construct a 3×4 matrix, whose elements are given by:

$$(i) \ a_{ij} = \frac{1}{2} |-3i + j| \qquad (ii) \ a_{ij} = 2i - j$$

6. Find the values of x , y and z from the following equations:

$$(i) \ \begin{bmatrix} 4 & 3 \\ x & 5 \end{bmatrix} = \begin{bmatrix} y & z \\ 1 & 5 \end{bmatrix} \qquad (ii) \ \begin{bmatrix} x+y & 2 \\ 5+z & xy \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 8 \end{bmatrix} \qquad (iii) \ \begin{bmatrix} x+y+z \\ x+z \\ y+z \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ 7 \end{bmatrix}$$

7. Find the value of a , b , c and d from the equation:

$$\begin{bmatrix} a-b & 2a+c \\ 2a-b & 3c+d \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 0 & 13 \end{bmatrix}$$

8. $A = [a_{ij}]_{m \times n}$ is a square matrix, if
 (A) $m < n$ (B) $m > n$ (C) $m = n$ (D) None of these
9. Which of the given values of x and y make the following pair of matrices equal

$$\begin{bmatrix} 3x+7 & 5 \\ y+1 & 2-3x \end{bmatrix}, \begin{bmatrix} 0 & y-2 \\ 8 & 4 \end{bmatrix}$$

 (A) $x = \frac{-1}{3}, y = 7$ (B) Not possible to find
 (C) $y = 7, x = \frac{-2}{3}$ (D) $x = \frac{-1}{3}, y = \frac{-2}{3}$
10. The number of all possible matrices of order 3×3 with each entry 0 or 1 is:
 (A) 27 (B) 18 (C) 81 (D) 512

3.4 Operations on Matrices

In this section, we shall introduce certain operations on matrices, namely, addition of matrices, multiplication of a matrix by a scalar, difference and multiplication of matrices.

3.4.1 Addition of matrices

Suppose Fatima has two factories at places A and B. Each factory produces sport shoes for boys and girls in three different price categories labelled 1, 2 and 3. The quantities produced by each factory are represented as matrices given below:

	Factory at A			Factory at B	
	Boys	Girls		Boys	Girls
1	80	60	1	90	50
2	75	65	2	70	55
3	90	85	3	75	75

Suppose Fatima wants to know the total production of sport shoes in each price category. Then the total production

In category 1 : for boys $(80 + 90)$, for girls $(60 + 50)$

In category 2 : for boys $(75 + 70)$, for girls $(65 + 55)$

In category 3 : for boys $(90 + 75)$, for girls $(85 + 75)$

This can be represented in the matrix form as
$$\begin{bmatrix} 80+90 & 60+50 \\ 75+70 & 65+55 \\ 90+75 & 85+75 \end{bmatrix}.$$